

Percent Applications

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS part = percent $\times$ whole. Turn the percent into a decimal first. Percent of a number: 20% of 50 = $0.20 \times 50 = 10$. Percent change = $(\text{new} - \text{old}) \div \text{old} \times 100$. Tax/tip/discount = price $\pm$ (percent $\times$ price).

Round 1 — Retrieve It

1. Write the formula that connects part, percent, and whole.

2. What do you do to a percent before multiplying?

3. Write the formula for percent change.

4. How do you find a final price after adding tax or a tip?

Round 2 — Sketch It

Sketch a bar model showing 20% of 50. Split the whole (50) into parts and shade the amount that represents 20%. Circle the value of the shaded part.

Percent Applications — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. What is 30% of 80?

2. A \$40 shirt is marked 25% off. What is the discount amount, and what is the sale price?

3. A price rises from 50 to 60. What is the percent change?

Round 4 — Reason from Evidence

A meal costs \$25. You add 8% tax and then leave a 15% tip on the original \$25. Find the tax, the tip, and the total amount paid. Show each step.

Percent Applications — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain why you change a percent to a decimal before multiplying. Use one example that is not printed on the cue card.

C

Connect

Connect discount and tax. Why can you use the same rule, percent $\times$ price, for both even though one lowers the price and one raises it?

E

Extend

Create your own percent-change problem. Give the numbers and show the work using (new $-$ old) $\div$ old $\times$ 100.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Percent Applications

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. $\text{part} = \text{percent} \times \text{whole}$.
2. Turn it into a decimal (e.g., $20\% = 0.20$).
3. $\text{percent change} = (\text{new} - \text{old}) \div \text{old} \times 100$.
4. Add the tax or tip to the price: $\text{final} = \text{price} + (\text{percent} \times \text{price})$.

Sketch It — look for

A bar for the whole (50) with 20% shaded; the shaded part equals $0.20 \times 50 = 10$, and that value (10) is circled.

Apply It

1. $0.30 \times 80 = \mathbf{24}$.
2. Discount = $0.25 \times 40 = \$10$; sale price = $40 - 10 = \mathbf{\$30}$.
3. $(60 - 50) \div 50 \times 100 = 10 \div 50 \times 100 = \mathbf{20\% \text{ increase}}$.

Reason from Evidence — look for

Tax = $0.08 \times 25 = \$2.00$. Tip = $0.15 \times 25 = \$3.75$. Total = $25 + 2.00 + 3.75 = \mathbf{\$30.75}$.

ACE — Extend look-for

Many answers possible. Check that the problem asks for percent change and the work correctly uses $(\text{new} - \text{old}) \div \text{old} \times 100$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.